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Technology Education Meets Invention Education: Towards Authentic Design and Problem Solving in School

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Abstract

Technology education is a global phenomenon that is a widespread and critical component of school systems in many countries. At the same time, different conceptions of the nature of technology education may be challenging for teachers and hamper authenticity and real-world connection. The (engineering) design process can be found as a backbone of technology education standards and curricula in many countries and is furthermore a common phenomenon that features across STEM subjects as well. In this position paper, therefore, the aim is to propose an infusion of elements from invention education into technology education, to more firmly ground subject matter specifically around design and thus increase authenticity of the subject and motivate students. Invention education is a novel educational intervention connected to STEM that explicitly foregrounds the processes and activities by which inventors approach problem solving, which are closely related to design. It is concluded that technology education may gain access to a more invention-oriented conceptual framework from exploring aspects of invention education which can, in turn, inform signature pedagogies in relation to design. At the same time, invention education may benefit from integrating with a well-established disciplinary tradition that exists in many countries throughout the world from exploring technology education more deeply.

Key Words: Invention Education, Technology Education, Design, Authenticity

1 INTRODUCTION

Technology education is a global phenomenon that is a widespread and critical component of school systems in many countries. At the same time, however, technology education also struggles for its place in the curriculum, not the least because of the great variation in how the subject is manifested in national curricula and standards (Jones et al., 2013; Nordlöf, 2022; Norström, 2022). In this regard, over the world technology education could be conceived of as based on or related to craft and sloyd (e.g., Finland), design (e.g., the UK, Norway), STEM (e.g., the USA), vocational education (e.g., the USA, South Africa), technological literacy (e.g., New Zealand, Sweden), or even be perceived as a digital subject concerned with digital tools and

systems, ICT, and computational thinking (e.g., Denmark, planned subject 2027, partly Australia, "Digital technologies") (ACARA, 2015; Gill et al., 2023; Norström, 2022). The different conceptions of the nature of technology education may be challenging for teachers since they lead to uncertainty about the subject matter to be taught, which in turn hampers authenticity and real-world connection of classrooms activities (Nordlöf, 2022).

There is, however, one element that is all but universal in technology education, namely, the (engineering) design process. It can be found as a backbone of standards and curricula in many countries and is furthermore a common phenomenon that features across STEM subjects as well (e.g., Hallström & Ankiewicz, 2023; Rossouw et al., 2011; Stables, 2020; Sung & Kelley, 2022). In this position paper, therefore, the aim is to propose an infusion of elements from invention education into technology education, to more firmly ground subject matter specifically around design and thus increase authenticity of the subject and motivate students. Invention education is a novel educational intervention connected to STEM that explicitly foregrounds the processes and activities by which inventors approach problem solving, which are closely related to design. Invention education has gained momentum in the USA and South Korea, for example, and could be defined both as a pedagogical approach and a field of research (Lemelson Foundation, forthcoming).

2 INVENTION EDUCATION: INVENTIVENESS AND PROBLEM SOLVING

At face value, invention education is an intuitive and easily conceived phenomenon; it is about experiences tied to learning about inventions, the invention process, entrepreneurship, and how to develop inventiveness and problem-solving skills. Though that may seem straightforward, the invention education community has not settled on a single definition so multiple conceptions abound. In this paper, we advance a conceptualization of invention education that builds upon the prior definition put forth by the Invention Education Research Group (2019):

> Invention education offers learners engagement in ideation, design, making, testing, revision and marketing of creations that they conceived to solve real-world problems of their choice. The main learning goal of IvE is occasionally the end product or invention. More often, the goal is on the process and the development of inventiveness—mindsets, capacities, skills and knowledge—frequently used by boundary-crossing inventors and valuable to the workforce of tomorrow. These educational opportunities may support learners through the entire process or select components and experiences that take place in multiple contexts over time with educators from various backgrounds facilitating the learning.

What this means in practice is something that researchers have enquired about without coming to a conclusive definition. In this section, we will outline a framework of strategies as a way of breaking down main components of invention education and connecting it to a broader technology education agenda.

2.1 Strategies for inventive problem solving

In recent work, Maltese et al. (2025) conducted a Delphi study with experts with roots in invention to identify the capacities and strategies necessary for doing inventive problem solving. This work produced a set of 20 capacities and 23 strategies that were identified as critical to problem solving. The list of *personal capacities* includes traits and abilities that an individual can develop that are helpful toward developing solutions, for example: 1) Seeking to identify with and share diverse perspectives, experiences, and/or feelings (Empathy), and 2) Adjusting goals and strategies to handle changing situations and circumstances (Adaptability). In contrast, *strategies* are characterized as approaches, heuristics or other components of problem solving that can be employed to assist in the process. Here we specifically focus on the 23 strategies as they can be easily compared with activities and components of the design process, see Table 1.

Table 1

Strategies for Inventive Problem Solving

Strategy	Description
Perspective taking	Examining problems and solutions from multiple points-of-view.
Critical thinking	Carefully evaluating information, evidence and options, through reasoning, reflection and analysis, to draw conclusions.
Visualization	Envisioning and considering potential outcomes and challenges before they arise.
Valuing diversity	Intentionally considering and including input from those with varied perspectives, backgrounds, and experiences.
Divergent thinking	Generating a wide range of ideas and possibilities related to a task or problem.
Inquiry	Active questioning, research, analysis, and comprehension; sometimes inclusive of experimentation and/or observation.
Design process	Engaging in cycles of idea generation, prototyping and testing to refine solutions or improve results.
Evaluation	Determining value or quality through use of evidence; may involve establishing and comparing against a set of criteria.
Synthesis	Making connections and consolidating ideas across existing knowledge, research and/or experiences.
Human-centered design	Considering and appropriately incorporating the needs of intended users/audience at all stages of the process.

Experimentation	Investigating a phenomenon through a designed set of tests, data collection, analysis and interpretation.
Questioning assumptions	Aware of and questioning biases, existing norms, assumptions, and processes.
Iteration	Engaging in multiple rounds of prototyping for the purpose of integrating new information or to improve outcomes.
Exploring complexity	Seeking to understand the relationships and dynamics between system components across domains.
Asset identification	Identifying and evaluating information, people and materials that may be useful toward solving a problem.
Observation	Noticing and discerning meaningful details, patterns, occurrences, and insights in information.
Reflection	Thinking deeply on the meaning or implications of experiences, beliefs and actions.
Incrementalism	An approach that involves accomplishing small goals toward a larger solution.
Problem definition	Defining and understanding the problem you're trying to solve, in detail.
Strategic thinking	Analyzing critical factors and variables that will influence long-term success.
Collaboration	Working with others, in an integrated way, toward a collective goal.
Leadership	Providing guidance to others in the form of oversight, direction, ideas and work to accomplish shared goals.
Communication	Bi-directional ability to convey/understand ideas, solutions, and perspectives presented through verbal, written and visual forms.

Note: Based on Maltese et al. (2025).

The strategies outlined in Table 1 are general and applicable to many different areas, and thus they constitute components of problem-solving capabilities or design thinking that may be useful while addressing any type of problem (e.g., Royalty, 2018). A similar case could be made for 21st century skills which also overlap with the strategies in Table 1, for example, creativity, innovation, critical thinking, and collaboration (Banks & Barlex, 2020). However, the strategies stem from a design and innovation context where each of them has an explicit meaning and role to play when through processes of design one invents and innovates products and systems that should meet customer desires and standards requirements (Glaa et al., 2023; Hallström & Schönborn, 2025; Maltese et al., 2025). Therefore, although they can be seen as general problem-solving strategies, they are, in fact, deeply rooted in (engineering) design, invention, and innovation, in all their variety. This makes sense given that inventions are often born from a desire

to solve an issue that an individual or group identifies or to improve aspects of life and work in some way.

3 TECHNOLOGY EDUCATION INFUSED WITH AN INVENTION EDUCATION PERSPECTIVE

By tradition, technology education has incorporated design as a central component, but at the same time the conception of design has often been formulaic and centred on one, linear model or step-based representation of the design process. Stables (2020) describes the challenges of relying on one "true" model of the design process as:

> a quest among professional designers to create a modernist, universal approach to designing, which was made manifest through a clinical, linear set of steps to be followed – a model that was adopted in schools and reinforced through assessment structures. Drawing on more recent design research, the linear model that persists as something often called 'the' design process has been challenged, questioning whether it is either true or helpful in developing design capability. […] This more recent analysis has provided a less simplistic, more authentic perspective, drawing together the speculative, wicked, uncertain, iterative nature of designing, negating an idea that there is one standard design process (p. 101).

Stables (2020) argues that pinning down with certainty exactly what is design knowledge might be construed as akin to the search for a holy grail, when design knowledge is more about having access to a repertoire of skills (p. 102). This more iterative and authentic notion of design, including key aspects about failure and iteration (e.g., Simpson & Maltese, 2017), is nowadays embraced both in technology education and invention education (e.g., Schut et al., 2020). However, although many of the above items from Maltese et al. (2025) are well known in technology, engineering, and design education contexts, the exact nature of their relation to design and the extent to which they are applied in curricula and classrooms remain unclear. In what follows we map central elements of invention education onto technology education with the objective of boosting design thinking and design processes for elementary and secondary classrooms. The selection of literature was performed following a protocol for narrative reviews, the object of which is to pinpoint pertinent literature for the topic at hand without following a pre-determined procedure (e.g., Demiris et al., 2019). Following an inductive, comparative reading of central literature, it was decided to compare Stables (2020) and Maltese et al. (2025) by mapping central facets of design in invention education onto signature design pedagogies in technology education (see Table 2). GenAI was used solely to soundboard initial mapping ideas.

Most, if not all, problem solving strategies in Table 1 relate to design directly or indirectly, but the extent to which they can be brought in as part of a repertoire of design thinking in technology education is hitherto unknown. Thus, in Table 2, we attempt to superimpose these strategies onto Stables' (2020) "signature pedagogies" (Shulman, 2005) of design knowledge, see Table 2.

Table 2

Signature Pedagogies for Design Knowledge and Strategies for Inventive Problem Solving

Signature Pedagogies (Stables, 2020)	Problem Solving Strategies (Maltese et al., 2025)
Pedagogies of speculation *Exploring possibilities, generating alternatives, asking "what might be?"*	Divergent thinking Exploring complexity Questioning assumptions Visualization
Pedagogies of imaging and modelling *Externalizing and refining ideas through representation, prototyping, and iterative design work*	Communication Design process Incrementalism Iteration Visualization
Pedagogies of need to know *Identifying and pursuing the knowledge needed to move a design forward*	Asset identification Inquiry Problem definition Exploring complexity
Pedagogies of critiquing *Evaluating ideas, reasoning through evidence, and reflecting on decisions*	Critical thinking Exploring complexity Evaluation Reflection Strategic thinking Synthesis

Pedagogies of collaboration *Designing with and for others; incorporating multiple perspectives*	Collaboration Communication Leadership Human-centered design Perspective taking Valuing diversity
Pedagogies of materiality *Learning through experimentation, observation, and interaction with materials or systems*	Asset identification Experimentation Observation

Note: Based on Stables (2020) and Maltese et al. (2025).

Table 2 presents our attempt to align two different sets of constructs that are related to teaching principles of design, among many. Whenever this type of comparison is done there are likely to be areas where the fit is better than others. Regardless of how perfect the fit is within and across these categories, we feel it is worth considering how such an alignment or reaching beyond disciplinary traditions may be beneficial to consider.

4 CONCLUDING DISCUSSION AND IMPLICATIONS

Just like invention education, it can be argued that technology education lacks a clear, unified knowledge base, manifested in the multitude of curricular representations that were exemplified in the beginning of the paper. Will we not just add a new layer and another set of concepts to an already shaky epistemological foundation? We believe not and thus present this paper to attempt to make explicit some linkages between disciplines and disciplinary practices that are typically not connected. We believe that the bridge between these two areas, invention education and technology education, can help the flow of ideas between these bodies of research literature and instructional practices and should enable multilateral improvements, epistemologically as well as in educational practice. We think that technology education may gain access to a more invention-oriented conceptual framework from exploring aspects of invention education (Table 1) which can, in turn, inform signature pedagogies in relation to design (Table 2). At the same time, invention education may benefit from integrating with a well-established disciplinary tradition that exists in many countries throughout the world from exploring technology education more deeply. For example, the signature pedagogies suggested by Stables (2020) could inform

invention education pedagogy with elements from a more structured conception of design that is at the same time focused on "wicked" problems and iterative problem-solving.

We thus argue that the signature pedagogies and identified strategies could lead to more effective problem-solving, inventiveness, and real-world relevance in technology education classrooms. Additionally, this will ground design and invention more clearly in the technology education subject philosophy, and vice versa, inform invention education with more structured philosophical and pedagogical elements from technology education. A downside is that the mapped pedagogies and strategies are conceptually very similar, so the mapping has been done with care and not as a 1:1 mapping but several strategies could be mapped onto more than one pedagogy. Apart from being a better mapping, it is, we believe, also more in line with the iterative nature of both design and invention.

Traditional presentations of the design process seem to be plagued by the same issues that are problematic in other disciplines when a non-linear process is distilled (cf., Keirl, 2020). For example, the "scientific method" is taught in schools around the world as a set of ordered steps. This method, which is an abstraction of science in action, may have some structure and temporality that makes sense in a school context, but if it is presented as a rigid set of steps it will not be representative of how science is done (e.g., Osborne et al., 2003; Rudolph, 2019), just like the notion of the design process and how it is carried out in practice. In a similar vein, invention education is not strictly about studying famous inventors or the patent process, just like science is not about only knowing established facts or doing laboratory work, or technology education is not merely about designing and making products. Although it is much easier for an educator to transmit concrete times, dates, places, facts, and procedures than taking time to develop students' creativity, critical thinking, or resilience, if the former is complemented with the latter it is obviously more authentic and beneficial to the long-term impact on our students, be it in science, invention, or technology education (e.g., Herrington et al., 2010).

As researchers, we may be interested in semantics or determining where fields overlap or not. Yet, the impact of this paper will likely be small if it is left at the stage presented here, where it is detached from the reality of education, educators, and learners. Thus, we argue there are some important next steps that we plan to pursue to push these ideas further. We plan to gather input from elementary and secondary educators in Sweden through interviews and observations of teaching, to understand how they conceive of this overlap and how they attempt to put some of these ideas into practice when teaching design-related and non-design content, in technology education and across STEM disciplines. We also plan to review the broader education ecosystem in Sweden to understand what current programming exists that provides youth with the opportunities to engage in these types of activities (e.g., design or invention competitions).

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